Nordic Flow: fix the allocation decision first, with a planner-approved service that cuts total logistics cost

Executive summary for the Board of Company A

THE PROBLEM	Goods allocation: which source and existing warehouse should serve each order. It sits where fulfilment, manual planning and fragmented systems meet.
WHO FEELS IT	Planners and warehouse teams in every market; customers feel it as late or split orders.
WHY IT MATTERS	Margins are thin, so cost-to-serve is a real lever. Outside studies report savings from about 5% (modelled) to 10–15% (realised) [cited].
WHAT WE PROPOSE	A planner-approved optimisation service on top of existing ERP, WMS and TMS. Nothing is replaced.
MVP	Sweden, one region: 2 suppliers, 3 warehouses, 20 SKUs, one weekly run. Eight weeks to a go / no-go [assumption].
WHERE AI HELPS	Data mapping, plain-language explanations, learning from planner overrides. The core is optimisation; a person approves.
HOW IT SCALES	Same engine with local data and rules: Sweden, then Denmark, then Norway, each wave gated by service and savings.

8 weeks
from approval to a go / no-go decision

≥ 5%
lower relevant logistics cost: our pilot gate [assumption]

3–6%
saving needed to break even in year 1 [assumption]

0
systems replaced; the planner approves every plan

Tags: [cited] = public source, listed on slides 2, 3, 8 and 9. [assumption] = replaced with Company A data in Week 1.

The allocation lever looks real but modest, so Week 1 measures it before we commit

CONFIRMED IN THE CASE

Hundreds of stores in Denmark, Norway and Sweden, each market with its own systems.

Fulfilment and fragmented systems affect efficiency and reliability.

Impact and urgency differ by market.

TO VERIFY IN WEEK 1

Planners optimise one leg at a time, not total cost.

Rates, stock and capacity sit in separate local systems.

Stock records may be wrong, so plans send goods to warehouses that do not have them.

WHAT OUTSIDE EVIDENCE SUGGESTS

4.8% lower delivered cost, building-products maker; modelled [cited]

10–15% logistics saving, 150+ warehouse network; realised, consultancy [cited]

65% of stock records inaccurate in one US retailer; profit down >10% [cited]

Byggmax 2025: *"restructured our logistics operations… increase control over flows" [cited]*

WHO FEELS IT Planners and warehouse teams (manual re-planning), branch staff (stock not where promised), customers (late or split orders).

WHY NOW Thin margins make cost-to-serve a lever: Byggmax EBITA 5.9% in 2025 [cited].

Reading: a few percent is plausible, not guaranteed. We gate the pilot on a measured cost gap, not on a promise.

Case brief pp. 8–11. David et al., Interfaces 45(3), 2015. McKinsey distribution-network case. DeHoratius & Raman, Management Science 54(4), 2008. Byggmax Annual Report 2025, p. 6 (EBITA 5.9%).

Goods allocation leads once impact, speed and data readiness are weighed together

	A Goods allocation	B Order processing	C Store standardisation	D Systems replacement	E Delivery visibility
Business impact 30%	4	3	3	5	5
Fit across DK / NO / SE 25%	4	3	2	5	3
Time to market 25%	4	4	2	1	3
Data readiness 20%	3	3	2	1	2
Weighted score	**3.80**	**3.25**	**2.30**	**3.20**	**3.40**

WHY A

An internal decision with no new driver, store or customer-facing change. It uses data the company already holds, so a CSV-based pilot runs in weeks.

ROBUSTNESS

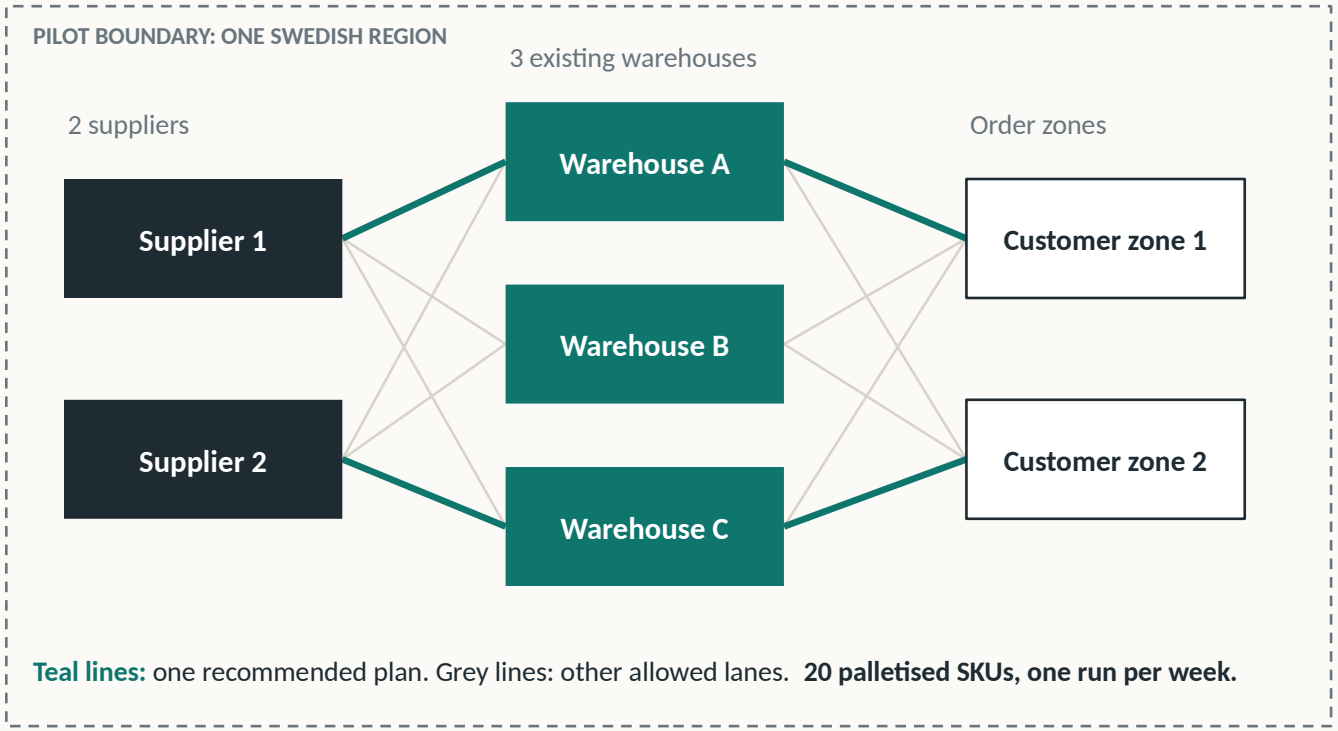
If impact is weighted 50%, E ties A at about 3.86 each. E is therefore our Phase 2: it reuses the same order and warehouse data.

NOT NOW

B helps mainly Denmark and its savings claims come from vendors. C needs broad store discovery. D is multi-year and high risk.

Scores 1–5 are team judgement against case evidence, not company data. Weights: impact 30%, fit across DK/NO/SE 25%, time to market 25%, data readiness 20%. Delivery visibility (E) added by the team.

A focused pilot: one Swedish region, three warehouses, 20 SKUs, one weekly planning run



Teal lines: one recommended plan. Grey lines: other allowed lanes. **20 palletised SKUs, one run per week.**

IN SCOPE

+ Allocate inbound supply and confirmed orders across existing warehouses
+ Compare the current allocation with a feasible lower-cost plan, with cost by leg
+ Planner approves; every override is logged

OUT OF SCOPE

– New warehouses or site selection
– Vehicle routing (stays in the current TMS)
– Demand forecasting, production planning
– ERP replacement; other countries

WHY SWEDEN FIRST The case ties Sweden to fulfilment impact. It is a discovery candidate, not a commitment: if Week 1 data is better in Denmark or Norway, the pilot market changes.

Sizes are [assumption]: the smallest network that still contains a real routing choice. Byggföretagen forecast: building investment +6% in total 2026–27, a fragile recovery [cited, autumn 2026 update]. Case p. 10 names delivery and fulfilment as Sweden's priority.

The planner stays in control: the service recommends, a person approves, existing systems execute

1 Load → **2 Compare** → **3 Review** → **4 Approve**

Orders, supply, rates and warehouse limits from existing systems

Current allocation versus a feasible lower-cost plan

Exceptions, service rules and cost by leg

Planner exports the plan; ERP, WMS and TMS execute it

RECOMMENDATION OUTPUT (EXAMPLE)

Supplier / SKU	Warehouse	Destination	Quantity / date
S1 / SKU-01	Warehouse A	Order zone North	60 pallets / within the week
S2 / SKU-01	Warehouse B	Order zone South	40 pallets / within the week

NOT IN THE PILOT

ERP replacement · automatic execution of plans · replacing planner judgement

Reuse before build: local systems stay authoritative. We add one thin layer of CSV / API adapters, a validation step, the optimiser and a planner view.

DATA THE PILOT NEEDS

Data	Lives in	Owner
Supply, orders	ERP	Planner
Rates, lanes	TMS, contracts	Logistics
Capacity, stock	WMS	Warehouse
Service rules	Local policy	Ops sponsor

Week 1 checks that each item exists and is usable.

Illustrative output only. Each run stores the input version, solver status, approvals and planner overrides.

A constrained optimisation engine does the maths; AI helps around it, always with a human approving

THE DECISION ENGINE

Minimise	inbound freight + handling + outbound freight + storage fees that change with the plan
Subject to	demand met, supply limits, warehouse handling limits, SKU eligibility, allowed lanes
Method	LP for divisible quantities; MILP for whole pallets and discrete commercial rules
Solver status	always shown: optimal, feasible within time limit, or infeasible with a diagnosis (short on supply, capacity or lanes)
Not counted	fixed rents already committed are not savings

WHERE AI ADDS VALUE

1 **Data mapping**

Suggests SKU and unit matches across country systems; a person validates.

2 **Plain-language explanations**

Why a warehouse changed and which constraint binds, so the planner can check quickly.

3 **Learning from overrides (Phase 2)**

Planner corrections refine the rules over time.

TECHNOLOGY Python service with open-source OR-Tools, EU-hosted, role-based access. A scenario store keeps input versions, solver status and approvals for audit.

RESPONSIBLE BY DESIGN Goods allocation is not an Annex III high-risk use [cited]. No profiling of staff. Data stays in EU hosting.

EU AI Act, Regulation (EU) 2024/1689, Art. 6 and Annex III. Google OR-Tools is open source (Apache 2.0).

A worked example shows where the saving comes from: seeing total cost across both legs

ROUTE COST PER PALLET (SOURCE → WAREHOUSE → ZONE)

	A / North	A / South	B / North	B / South
S1	**5**	9	11	8
S2	6	10	8	**5**

Teal: optimised plan. **Amber outline:** baseline route that the optimiser changes.

Supply S1 / S2 = 60 / 40. Demand North / South = 60 / 40. Warehouse capacity A / B = 100 / 60. Both plans serve all 100 pallets.

TOTAL COST (COST UNITS)

Baseline: always use preferred warehouse A



700

Optimised: send the second 40 pallets via warehouse B



500

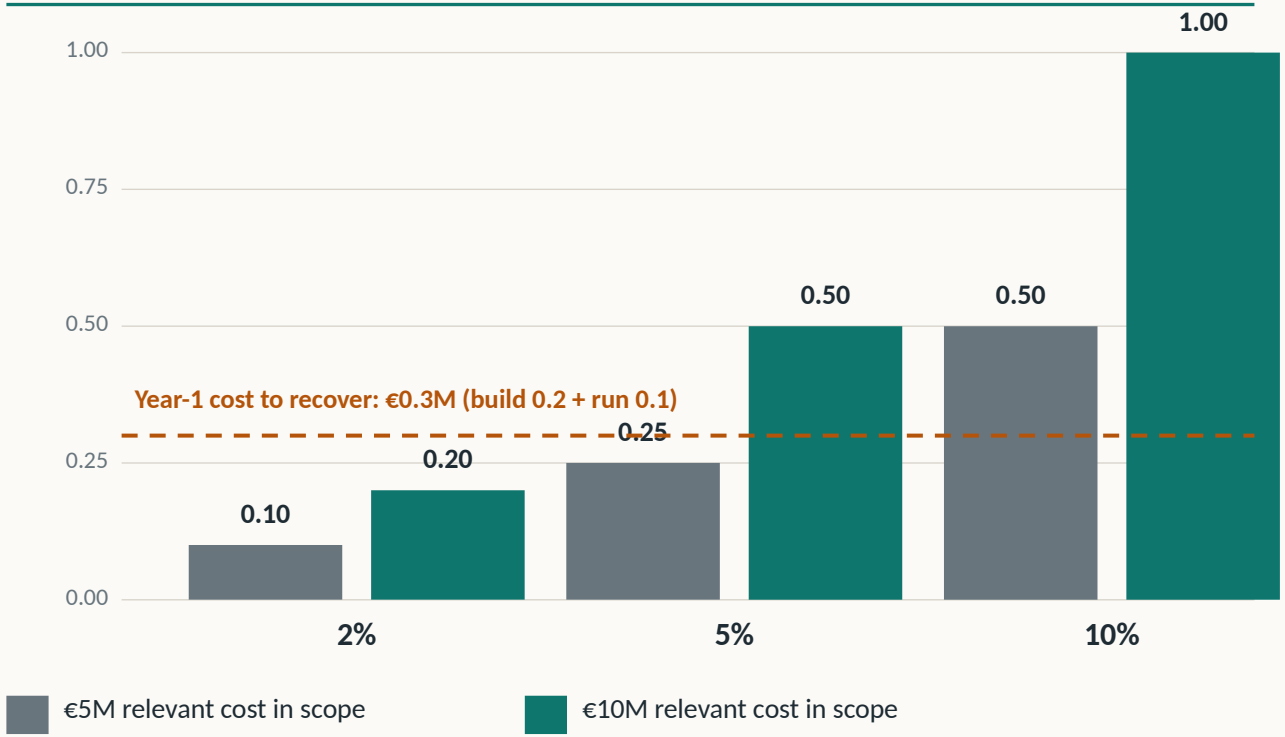
Saving: 200 units, 28.6%, on synthetic data only.

What it shows. The saving comes from seeing both legs together, not from a cheaper single leg. Habit fills the preferred warehouse; the optimiser uses the cheaper route that still respects capacity. Real gains are expected to be far smaller than 28.6%; the next slide shows what must be true.

Synthetic data for illustration: 1 SKU, 2 source warehouses, 2 destination zones. It demonstrates the mechanism and is not a forecast for Company A.

The case to proceed: a saving of only 3–6% breaks even in year one, and Week 1 measures the real figure

ANNUAL SAVING BY SAVING RATE (€ MILLION)



Break-even rate = €0.3M ÷ relevant cost: **6.0%** at €5M, **3.0%** at €10M. The 5% gate clears break-even when the relevant cost is about €6M or more.

PILOT ACCEPTANCE GATES (HYPOTHESES)

Relevant logistics cost

≥ 5% below a validated feasible baseline, same orders

Planner preparation time

≥ 30% lower, full task including data checks

Service

No deterioration in on-time, in-full rate

Feasibility

Zero hard-constraint violations

Scale only when validated savings cover delivery and operating cost. No revenue uplift is assumed.

All cost inputs are [assumption], replaced with Company A data in Week 1. Outside benchmarks: ~4.8% modelled (Interfaces, 2015), 10–15% realised (McKinsey). Only costs that change with allocation count; labour savings are assessed separately.

One engine, three rollouts: each market gets the same core with local data and rules

WAVE	LOCAL EMPHASIS	EVIDENCE TO COLLECT FIRST
1	**Sweden** Pilot one region from CSV exports; the planner reviews every plan.	Rates, stock, service rules and warehouse capacity.
2	**Denmark** Reuse the engine; adapt the import to cut manual preparation.	Structured B2B e-invoices only 5–10% [cited], so inputs need more cleaning. Wholesale EDI sales 39% [cited].
3	**Norway** Validate branch and distribution workflows before connecting.	Structured B2B e-invoices 89% [cited]; wholesale EDI sales 27% [cited]. Store receiving limits, carrier rules.

Shared core: data contract · optimisation rules · audit trail · KPI definitions. Local teams own rates, service rules and approval rights.

EMPLOYEES

Planners co-design the workflow and keep the final say; overrides are logged, not punished. Training in the first week of each wave.

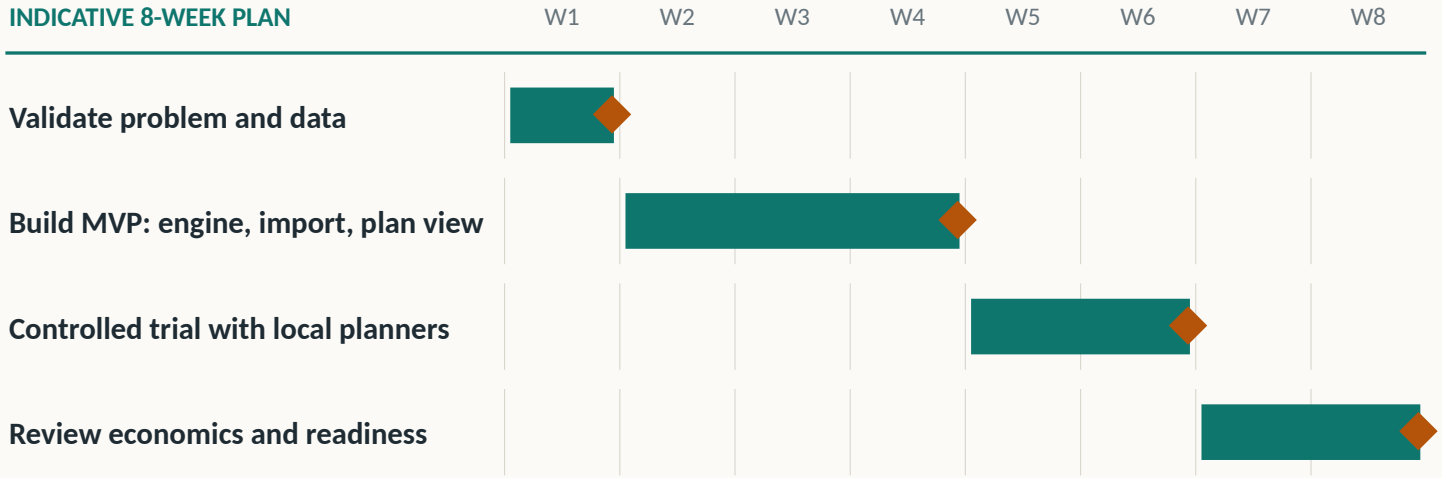
CUSTOMERS

No change to ordering. The on-time, in-full rate is a gate before each wave, so service cannot quietly worsen.

Eurostat isoc_ec_eseln2 (2024, wholesale, ≥10 employees); Digdir (NO) and Erhvervsstyrelsen (DK) e-invoicing analyses. Case p. 11: one strategic solution does not mean one identical implementation. Rollout order is a proposal.

Eight weeks to a pilot decision, with the main risks designed out and a clear ask

INDICATIVE 8-WEEK PLAN



◆ **Exit gates:** agreed baseline · replay passed · planner acceptance · go / revise / stop

MAIN RISKS

Stock or rate data is wrong	65% of records in one study. Validation step; block invalid runs; planner approves.
Inbound freight paid by suppliers	Seen at Byggmax. Week 1 confirms who pays; if suppliers, optimise the warehouse-to-customer leg and handling only.
Weak baseline overstates the gain	Compare against a validated feasible baseline on the same orders.
Low planner adoption	Co-design, keep overrides, show cost by leg.

DECISION REQUESTED

1 Approve Week 1 Swedish discovery and a capped pilot design.

2 Name a business owner and a local process owner.

3 Proceed only if data, a measured cost gap and ownership are confirmed; otherwise iterate or stop.

Stock-record accuracy: DeHoratius & Raman (2008). Plan and sizes are indicative and depend on data access.